Hydrogen Atom

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The radial part of the ground state of the hydrogen atom is taken as

$$R_{10}(r) = A\, e^{-r/a_0}, \tag{1}$$

where A is the normalization constant and a_0 is the Bohr radius.

1. **Normalization, radial probability, and most probable distance.**
   (a) Determine the normalization constant A .
   (b) Plot the *radial probability density vs distance r.*
   (c) Find the most probable radius r_{mp}.

2. **Probability within the Bohr radius.**
   Compute $\Pr(r < a_0)$.

3. **Expectation values.**
   Evaluate the expectation values

   $$\langle r \rangle, \qquad \langle r^2 \rangle, \qquad \left\langle \frac{1}{r} \right\rangle$$

   for the ground state, and report the ratio $\dfrac{r_{\text{mp}}}{\langle r \rangle}$.

4. Calculate the momentum uncertainty in the ground state. The momentum operator is given by,

   $$p_r = -i\hbar\left(\frac{\delta}{\delta r} + \frac{1}{r}\right)$$